

CALCULUS WTW158
EXAMINATION / EKSAMEN

Date / Datum: June / Junie 2014 Marks / Punte : 55 Time / Tyd : 3 hours / uur

MARKS FOR SECTION B OUT OF 31 PUNTE VIR AFDELING B UIT 31	
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SURNAME / VAN

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FIRST NAMES / VOORNAME

STUDENT NUMBER

STUDENTENOMMER

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SIGNATURE

HANDTEKENING

Internal Examiners / Interne Eksaminatore	WTW 158 Lecturers
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External Examiner / Eksterne Eksaminator	Ms / Me R Möller
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READ THE FOLLOWING INSTRUCTIONS

1. The paper consists of pages 1 to 13. Check whether your paper is complete.
2. No calculator may be used.
3. No question paper may be removed from the venue.

LEES DIE VOLGENDE INSTRUKSIES

1. Die vraestel bestaan uit bladsye 1 tot 13. Kontroleer of u vraestel volledig is.
2. Geen sakrekenaar mag gebruik word nie.
3. Geen vraestel mag uit die lokaal geneem word nie.

SECTION A: MULTIPLE CHOICE QUESTIONS - 24 MARKS: $1\frac{1}{2}$ MARKS EACH

READ THE FOLLOWING INSTRUCTIONS

1. YOU HAVE TO START WITH SECTION A, AS YOU HAVE TO HAND IN THE OPTIC READER FORM AFTER 75 MINUTES.
2. As you may not erase wrong answers on the optic reader form, first circle your answers on this paper and only mark the answers on the form when you are certain of your choice.
3. Do all scribbling on the facing page. It will not be marked.
4. Use a soft pencil to write on the optic reader form.
5. Use **SIDE 2** of the optic reader form.
Fill in your personal information in pencil on side 2.
Fill in your student number from top to bottom and then code it.
6. You may start with section B as soon as you have marked your answers on the optic reader form.

AFDELING A : MEERVOUDIGE KEUSEVRAE - 24 PUNTE: $1\frac{1}{2}$ PUNTE ELK

LEES DIE VOLGENDE INSTRUKSIES

1. U MOET MET AFDELING A BEGIN WANT U MOET DIE MERKLEESVORM NA 75 MINUTE INHANDIG.
2. U mag nie verkeerde antwoorde uitvee op die merkleesvorm nie. Omkring dus u antwoorde eers in hierdie vraestel en merk die antwoorde op die vorm as u seker is van u keuse.
3. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
4. Gebruik 'n sagte potlood om op die merkleesvorm te skryf.
5. Gebruik **KANT 2** van die merkleesvorm.
Vul u persoonlike inligting in potlood in op kant 2.
Vul u studentenommer van bo na onder in en kodeer dit.
6. U kan met afdeling B begin sodra u die antwoorde op die merkleesvorm ingevul het.

Question 1 / Vraag 1

Assume that f and g are differentiable functions of which the following is known:

Neem aan dat f en g differensieerbare funksies is, waarvan die volgende bekend is:

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	3	1	4	6

Let / Laat $p(x) = \sqrt{f(x)g(x) + x^2}$. Then / Dan is $p'(1) =$

(a)	0	(b)	48	(c)	$6\frac{1}{2}$	(d)	$\frac{1}{2\sqrt{26}}$
(e)	$5\frac{1}{2}$	(f)	6	(g)	None of these / Geen van hierdie		

The information below applies to Questions 2 to 4

Die onderstaande inligting is van toepassing op Vrae 2 tot 4

Consider the function $f(x) = \frac{x^2}{(x+1)^2}$ with its derivative $f'(x) = \frac{2x}{(x+1)^3}$.

Beskou die funksie $f(x) = \frac{x^2}{(x+1)^2}$ met sy afgeleide $f'(x) = \frac{2x}{(x+1)^3}$.

Question 2 / Vraag 2

The equation(s) of the horizontal asymptote(s) are given by

Die vergelyking(s) van die horisontale asimptoot(-tote) word gegee deur

(a)	$y = 0$	(b)	$y = 0; y = -1$	☒ (c)	$y = 1$	(d)	$y = \pm 1$
(e)	f has no horizontal asymptotes f het geen horisontale asimptote nie			(f)	None of these / Geen van hierdie		

Question 3 / Vraag 3

The equation(s) of the vertical asymptote(s) are given by

Die vergelyking(s) van die vertikale asimptoot(-tote) word gegee deur

(a)	$x = 0$	(b)	$x = 0; x = -1$	☒ (c)	$x = -1$	(d)	$x = 0; x = 1$
(e)	f has no vertical asymptotes f het geen vertikale asimptote nie			(f)	None of these / Geen van hierdie		

Question 4 / Vraag 4

The interval(s) on which the function f is increasing is

Die interval(le) waarop die funksie f stygend is, is

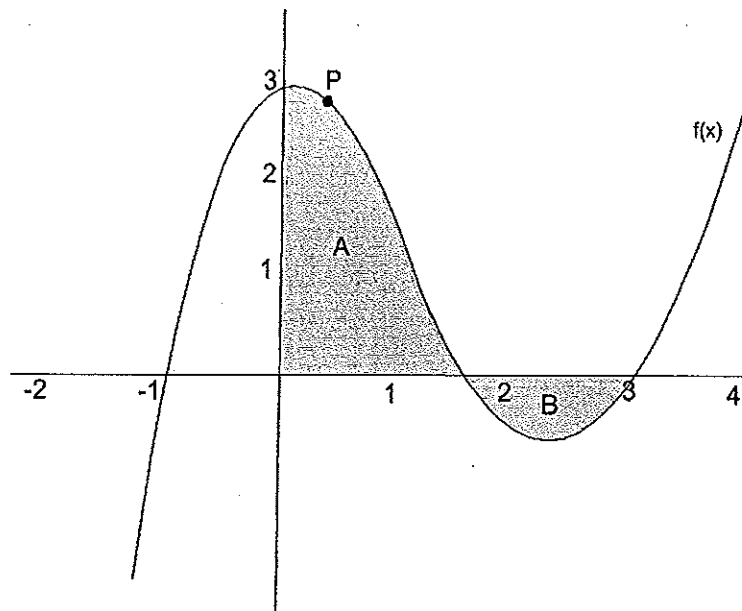
(a)	$(-1, 0)$	☒ (b)	$(-\infty, -1) \cup (0, \infty)$	(c)	$(-\infty, -2) \cup (-1, \infty)$	(d)	$\mathbb{R}$
(e)	f decreases on $\mathbb{R}$ f daal op $\mathbb{R}$		(f)	$(-3, 0) \cup (0, -1)$	(g)	None of these / Geen van hierdie	

The information below applies to Questions 5 to 8

Die onderstaande inligting is van toepassing op Vrae 5 tot 8

Given the graph of a function $f(x)$, where A and B indicate the areas of the shaded regions and P the point $(0.4, 2.8)$.

Gegee die grafiek van 'n funksie $f(x)$, waar A en B die oppervlakte van die gearseerde gebiede aandui en P die punt $(0.4, 2.8)$.



Question 5 / Vraag 5

Let $g(x) = -\frac{1}{3}f(x) + 1$. The y -intercept of g is

Laat $g(x) = -\frac{1}{3}f(x) + 1$. Die y -afsnit van g is

(a)	$y = -2$	(b)	$y = -1$	(c)	$y = 1$
(d)	$y = 0$	(f)	None of these / Geen van hierdie		

Question 6 / Vraag 6

Which one of the following statements is true?

Watter een van die volgende bewerings is waar?

(a) $f'(0.4) < 0$ and / en $f''(0.4) < 0$	(b) $f'(0.4) < 0$ and / en $f''(0.4) > 0$
(c) $f'(0.4) > 0$ and / en $f''(0.4) < 0$	(d) $f'(0.4) > 0$ and / en $f''(0.4) > 0$
(e) None of these / Geen van hierdie	

Question 7 / Vraag 7

If $C = \int_0^3 f(x) dx$, then / As $C = \int_0^3 f(x) dx$, dan is

(a) $C = A - B$	(b) $C = -A + B$	(c) $C = A + B$	(d) $C = -A - B$
(e) None of these / Geen van hierdie			

Question 8 / Vraag 8

Let k be a function with $k'(x) = f(x)$. Which one of the following statement(s) is true?

Laat k 'n funksie wees met $k'(x) = f(x)$. Watter een van die volgende bewering(s) is waar?

(a) k has a local maximum at $x = 3 / k$ het 'n lokale maksimum by $x = 3$
☒ (b) k has a local minimum at $x = 3 / k$ het 'n lokale minimum by $x = 3$
(c) k has a point of inflection at $x = 3 / k$ het 'n buigpunt by $x = 3$
(d) k has a zero at $x = 3 / k$ het 'n nulpunt by $x = 3$
(e) None of these / Geen van hierdie

Question 9 / Vraag 9

The domain, D , and the range, W , of $f(x) = \sin^{-1}(2x - 1)$ are given by

The definisieversameling, D , en die waardeversameling, W , van $f(x) = \sin^{-1}(2x - 1)$ word gegee deur

(Notation / Notasie: $\sin^{-1} = \text{bgsin} x = \arcsin$)

(a) $D = [0, 1]$ and/en $W = (-\infty, \infty)$	(b) $D = [-\frac{\pi}{2}, \frac{\pi}{2}]$ and/en $W = [0, 1]$
☒ (c) $D = [0, 1]$ and/en $W = [-\frac{\pi}{2}, \frac{\pi}{2}]$	(d) None of these / Geen van hierdie

Question 10 / Vraag 10

Consider the function $f(x) = \sin x + \cos x$ on the interval $[0, 2\pi]$. The x -value(s) of the point(s) where the tangent line(s) to the graph of $f(x)$ is (are) horizontal is (are)

Beskou die funksie $f(x) = \sin x + \cos x$ op die interval $[0, 2\pi]$. Die x -waarde(s) van die punt(e) waar die raaklyn(e) aan die grafiek van $f(x)$ horisontaal is, is

(a) $\frac{5\pi}{4}$ and/en $\frac{7\pi}{4}$	☒ (b) $\frac{\pi}{4}$ and/en $\frac{5\pi}{4}$
(c) $\frac{\pi}{4}$ and/en $\frac{7\pi}{4}$	(d) None of these / Geen van hierdie

Question 11 / Vraag 11

$\lim_{h \rightarrow 0} \frac{\sqrt[4]{16+h} - 2}{h}$ represents the derivative of a function f at the point $x = a$ with

$\lim_{h \rightarrow 0} \frac{\sqrt[4]{16+h} - 2}{h}$ stel die afgeleide van 'n funksie f by die punt $x = a$ voor met

(a) $f(x) = \sqrt[4]{x}, a = 2$	(b) $f(x) = \sqrt[4]{16+x}, a = 4$	(c) $f(x) = \sqrt[4]{16+x}, a = 16$
☒ (d) $f(x) = \sqrt[4]{x}, a = 16$	(e) None of these / Geen van hierdie	

Question 12 / Vraag 12

$$\int_0^4 (x + \sqrt{16 - x^2}) dx =$$

(a)	$4 + 8\pi$	(b)	$4 + 4\pi$	(c)	$8 + 4\pi$
(d)	$8 + 16\pi$	(e)	None of these / Geen van hierdie		

Question 13 / Vraag 13

$$\lim_{x \rightarrow -\infty} \frac{1}{\tan^{-1}(2x)} = \quad (\text{Notation: } \tan^{-1} = \arctan / \text{Notasie: } \tan^{-1} = \text{bgtan})$$

(a)	$\frac{\pi}{2}$	(b)	$-\frac{2}{\pi}$	(c)	$\frac{2}{\pi}$
(d)	$-\infty$	(e)	None of these / Geen van hierdie		

Question 14 / Vraag 14

$$\vec{a} \cdot \vec{b} = (\vec{i} + 2\vec{j} + 3\vec{k}) \cdot (-\vec{i} - 2\vec{j} - 3\vec{k}) =$$

(a)	$\vec{i} + 2\vec{j} + 3\vec{k}$	(b)	$-\vec{i} - 2\vec{j} - 3\vec{k}$	(c)	-14	(d)	14
(e)	-11	(f)	$-\vec{i}$	(g)	None of these / Geen van hierdie		

Question 15 / Vraag 15

A vector orthogonal to both $\vec{a} = \langle 2, 0, -1 \rangle$ and $\vec{b} = \langle 1, 4, 3 \rangle$, is

'n Vektor ortogonaal op beide $\vec{a} = \langle 2, 0, -1 \rangle$ en $\vec{b} = \langle 1, 4, 3 \rangle$, is

(a)	$\langle 4, 7, 8 \rangle$	(b)	$\langle 2, 0, -3 \rangle$	(c)	$\langle 1, 0, 2 \rangle$
(d)	$\langle 4, -7, 8 \rangle$	(e)	None of these / Geen van hierdie		

Question 16 / Vraag 16

If f is a continuous function on $[a, b]$, which of the following statements is always true?

As f 'n kontinue funksie op $[a, b]$ is, watter van die volgende stellings is altyd waar?

(a)	f' exists on (a, b) / f' bestaan op (a, b)
(b)	If $f(x_0)$ is a maximum of f , then $f'(x_0) = 0$ / As $f(x_0)$ 'n maksimum is van f , dan is $f'(x_0) = 0$
(c)	then $\int_a^b f(x) dx$ exists / dan bestaan $\int_a^b f(x) dx$
(d)	$f'(x) = 0$ for some $x \in [a, b]$ / $f'(x) = 0$ vir een of ander $x \in [a, b]$
(e)	The graph of f' is a straight line / Die grafiek van f' is 'n reguit lyn

SECTION B: 31 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. Answer all the following questions on this paper in ink.
2. Work done in pencil or in red ink will not be marked.
3. You are not allowed to use correction fluid ("Tipp-Ex").
4. Do all scribbling on the facing page. It will not be marked.
If you need more space for an answer, use the facing page and indicate it clearly by framing it.
5. Show all steps and calculations and explain your answers.

AFDELING B: 31 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. Beantwoord die volgende vrae op hierdie vraestel in ink.
2. Werk wat in potlood of rooi ink beantwoord is, word nie nagesien nie.
3. U mag nie korrigeer-vloeistof ("Tipp-Ex") gebruik nie.
4. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
As u nog ruimte vir 'n antwoord nodig het, gebruik die teenblad en dui dit duidelik aan deur 'n raam daarom te trek.
5. Toon alle stappe en berekeninge en verduidelik u antwoorde.

Question 1 / Vraag 1

If / As $f''(x) = \frac{1}{x^3}$, $f(1) = 3$ and / en $f'(1) = -1$, determine / bepaal $f(x)$.

$$f'(x) = \frac{x^{-2}}{-2} + C_1 = -\frac{1}{2x^2} + C_1$$

$$f(x) = \frac{x^{-1}}{-1} + C_1 x + C_2 = -\frac{1}{2x} + C_1 x + C_2$$

$$f'(1) = -\frac{1}{2} + C_1 = -1$$

$$\Rightarrow C_1 = -\frac{1}{2} \quad \text{and}$$

$$f(1) = -\frac{1}{2} + C_1(1) + C_2 = 3$$

$$\Rightarrow -\frac{1}{2} - \frac{1}{2} + C_2 = 3$$

$$C_2 = 3$$

$$\Rightarrow f(x) = -\frac{1}{2x} - \frac{1}{2}x + 3$$

Question 2 / Vraag 2

Determine the derivatives of the following functions. (Do not simplify your answers.)

Bepaal die afgeleides van die volgende funksies. (Moenie jou antwoorde vereenvoudig nie.)

2.1 $f(x) = 2^{x^2}$

$$f'(x) = 2^{x^2} \cdot \ln 2 \cdot 2x$$

1

2.2 $h(t) = t \sin\left(\frac{1}{t}\right)$

$$h'(t) = \sin\left(\frac{1}{t}\right) + t \cos\left(\frac{1}{t}\right) \cdot -\frac{1}{t^2}$$

2

2.3 $f(x) = \frac{1}{\sin^{-1}(2x)}$ (Notation / Notasie: $\sin^{-1} = \arcsin$).

$$f'(x) = \frac{-1}{[\sin^{-1}(2x)]^2} \cdot \frac{2}{\sqrt{1-4x^2}}$$

2

2.4 $f(x) = (x)^{\tan x}$

Let $y = x^{\tan x}$

$$\Rightarrow \ln y = \tan x \ln x$$

$$\Rightarrow \frac{y'}{y} = \sec^2 x \cdot \ln x + \tan x \cdot \frac{1}{x}$$

$$\Rightarrow y' = y \left[\sec^2 x \cdot \ln x + \frac{\tan x}{x} \right]$$

2

Question 3 / Vraag 3

Determine the following integrals:

Bepaal die volgende integrale:

$$\begin{aligned} 3.1 \int \frac{3x-1}{1+x^2} dx &= \int \left(\frac{3x}{1+x^2} - \frac{1}{1+x^2} \right) dx \\ &= \frac{3}{2} \ln(1+x^2) - \arctan x + C \end{aligned}$$

2

$$\begin{aligned} 3.2 \int \frac{e^{\cos x}}{\operatorname{cosec} x} dx &= \int e^{\cos x} \cdot \sin x dx \\ &= -e^{\cos x} + C \end{aligned}$$

1

$$\begin{aligned} 3.3 \int \frac{e^{3x}}{\sqrt{1-e^{6x}}} dx &= \int \frac{e^{3x}}{\sqrt{1-(e^{3x})^2}} dx \\ &= \frac{1}{3} \arcsin e^{3x} + C \end{aligned}$$

1

$$3.4 \int_0^{\pi/4} \tan 3x \, dx$$

$$= \int_0^{\pi/4} \frac{\sin 3x}{\cos 3x} \, dx$$

$$= -\frac{1}{3} \ln |\cos 3x| \Big|_0^{\pi/4}$$

$$= -\frac{1}{3} \ln \left| -\frac{1}{\sqrt{2}} \right| + \frac{1}{3} \ln |1|$$

$$= -\frac{1}{3} \ln \frac{1}{\sqrt{2}} = \frac{1}{3} \ln \sqrt{2} = \frac{1}{6} \ln 2$$

2

$$3.5 \int \frac{x}{\sqrt{1+3x^2}} \, dx$$

$$= \int x (1+3x^2)^{-1/2} \, dx$$

$$= \frac{1}{6} \cdot 2 (1+3x^2)^{1/2} + C$$

$$= \frac{1}{3} \sqrt{1+3x^2} + C$$

1

$$3.6 \int \sec(a^{t^2}) \tan(a^{t^2}) t a^{t^2} \, dt$$

$$= \frac{1}{2 \ln a} \sec(a^{t^2})$$

1

Question 4 / Vraag 4

Determine the following limits, if they exist. / Bepaal die volgende limiete, indien hulle bestaan.

4.1 $\lim_{x \rightarrow 1^+} (x)^{1/(1-x)}$

let $y = x^{\frac{1}{1-x}}$

$$\Rightarrow \ln y = \frac{1}{1-x} \ln x$$

$$\begin{aligned} \Rightarrow \lim_{x \rightarrow 1^+} \ln y &= \lim_{x \rightarrow 1^+} \frac{\ln x}{1-x} && \left(\frac{0}{0}\right) \\ &= \lim_{x \rightarrow 1^+} \frac{\frac{1}{x}}{-1} \\ &= -1 \end{aligned}$$

$$\Rightarrow \lim_{x \rightarrow 1^+} y = e^{-1} = \frac{1}{e}.$$

3

4.2 $\lim_{x \rightarrow 0.5^-} \frac{|2x-1|}{4x^2-1}$

$$|2x-1| = \begin{cases} 2x-1 & \text{if } x \geq \frac{1}{2} \\ -(2x-1) & \text{if } x < \frac{1}{2} \end{cases}$$

$$= \lim_{x \rightarrow 0.5^-} \frac{-(2x-1)}{4x^2-1}$$

$$= \lim_{x \rightarrow 0.5^-} \frac{-1}{2x+1}$$

$$= -\frac{1}{2}$$

2

Question 5 / Vraag 5

Find the interval(s) on which the function $f(x) = x^2 \ln x$ is increasing.
Bepaal die interval(le) waarop die funksie $f(x) = x^2 \ln x$ stygend is.

Domain of
 f is $(0, +\infty)$

$$f'(x) = 2x \ln x + x^2 \cdot \frac{1}{x}$$

f increasing where

$$2x \ln x + x > 0$$

$$\Rightarrow x(2 \ln x + 1) > 0$$

$$\Rightarrow 2 \ln x + 1 > 0 \text{ since } x > 0$$

$$\Rightarrow \ln x > -\frac{1}{2}$$

$$\Rightarrow x > e^{-1/2}$$

f increases on $(e^{-1/2}, +\infty)$

Question 6 / Vraag 6

Given / Gegee $f(x) = \cos x$, $x \in (0, \frac{3\pi}{2})$.

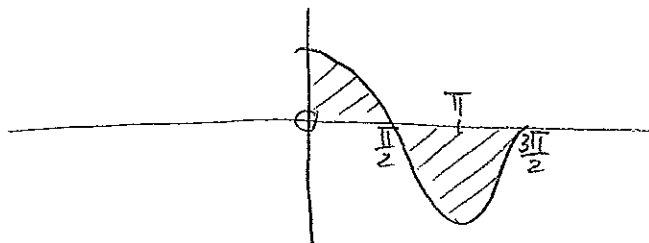
6.1 Evaluate / Bepaal

$$\int_0^{\frac{3\pi}{2}} f(x) dx$$

$$= \sin x \Big|_0^{\frac{3\pi}{2}} = -1$$

6.2 Sketch the graph of f and determine the area between the curve of the function f , the x -axis and the lines $x = 0$ and $x = \frac{3\pi}{2}$.

Skets die grafiek van f en bepaal die oppervlakte tussen die kromme van f , die x -as en die lyne $x = 0$ en $x = \frac{3\pi}{2}$.



$$A_1 = \int_0^{\frac{\pi}{2}} \cos x dx$$
$$= \sin x \Big|_0^{\frac{\pi}{2}} = 1$$

$$A_2 = \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \cos x dx$$
$$= \sin x \Big|_{\frac{\pi}{2}}^{\frac{3\pi}{2}} = -1 - 1 = -2$$

$$\Rightarrow \text{Area} = 1 + 2 = 3$$